

Computer Networks (CS 3001)

Final Exam

Total Time (min): 180
Total Marks: 100
Total Questions: 5

Date: 31st May , 2024

Course Instructor(s)

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Roll No

Section

Student Signature

Do not write below this line.

Attempt all the questions.

CLO # 1: Describe and evaluate the protocols, services and functions provided by each layer in the Internet protocol stack

Q1.a) Answer the following questions in two to three sentences:

- i. What do you mean by denial of service attack? **(1)**
- ii. What do you mean by rarest first in BitTorrent Networks?(2)
- iii. If a client wishes to access www.amazon.com. How this request will be processed by the hierarchy of DNS servers? **(2)**

b) Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps. **(2.5+2.5)**

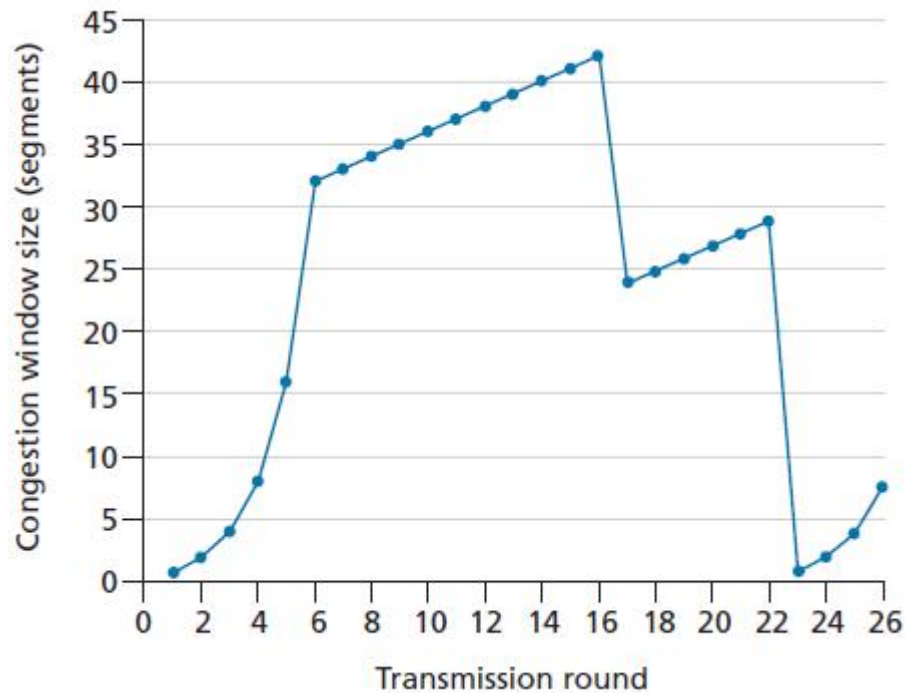
- i) Assuming no other traffic in the network, what is the throughput for the file transfer?
- ii) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?

c) Suppose users share a 10 Mbps link. Also suppose each user requires 200 kbps when transmitting, but each user transmits only 10 percent of the time. When circuit switching is used, how many users can be supported? **(5)**

CLO #2: Apply network protocol and communication services for client/server and other application layouts

Q2. a) Consider the following figure, assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. (10)

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- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of ssthresh at the first transmission round?

b) Suppose that the five measured SampleRTT values are 106 ms, 120 ms, 140 ms, 90ms and 115ms. Compute the EstimatedRTT after each of these SampleRTT values is obtained, using a value of $\alpha = 0.125$ and assuming that the value of Estimated RTT was 100 ms just before the first of these five samples were obtained. Compute also the DevRTT after each sample is obtained, assuming a value of $\beta = 0.25$ and assuming the value of DevRTT was 5 ms just before the first of these five samples were obtained. Last, compute the TCP TimeoutInterval after each of these samples is obtained. (10)

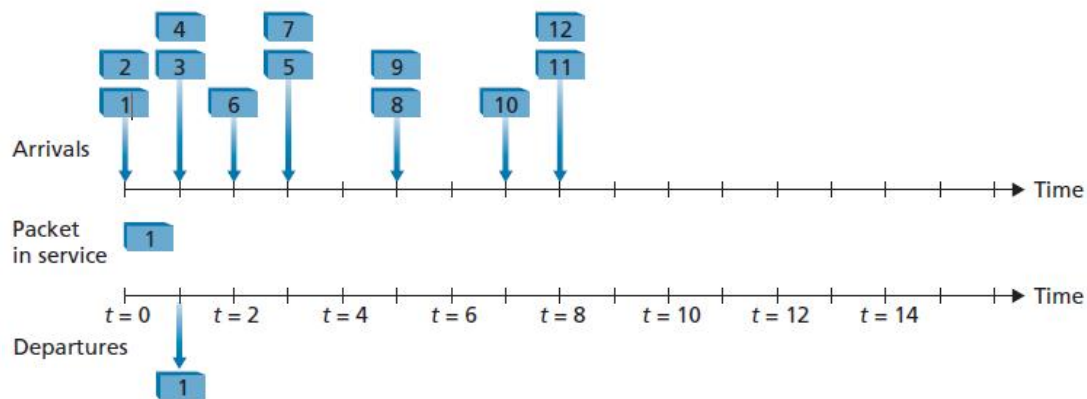
CLO #3: Analyze the architectural principles of computer networking and compare different approaches to organizing networks.

Q3.a) Describe the concept of Software Defined Networks? How can we differentiate it with traditional network? (5)

b) You need to schedule network packets using Priority service. Assume that odd-numbered packets are high priority, and even-numbered packets are low priority. Indicate the time at which packets 1 through 12 each leave the queue. For each packet, what is the delay between its arrival and the beginning of the slot in which it is transmitted? What is the average of this delay over all 12 packets? (5)

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c) Suppose datagrams are limited to 1,500 bytes (including header) between source Host A and destination Host B. Assuming a 20-byte IP header, how many datagrams would be required to send an MP3 consisting of 74,000bytes? **(5)**

d) Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3 and packets are to be forwarded to the link interfaces as follows: **(2.5+2.5)**

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

i. Suppose a datagram arrives at the router, with destination address 11001000 10010001 01010001 01010101. To which interface will this datagram be forwarded using longest-prefix matching?

ii. Suppose a datagram arrives at the router, with destination address 11100001 01000000 11000011 00111100. To which interface will this datagram be forwarded using longest-prefix matching?

e) IPV4 supports Variable Length Subnet Mask for designing a network. You need to calculate subnet bits, host per subnet, and subnet mask of Table 1.

(15)

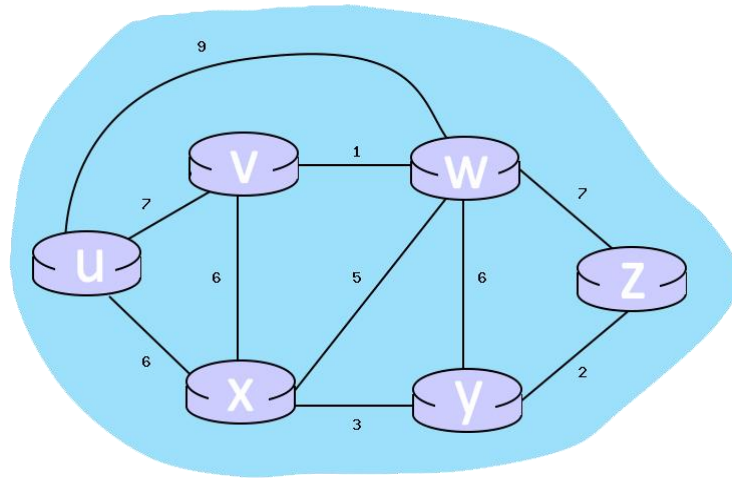
Networks	Subnet Bits	Host / Subnet	Subnet Mask
10.0.0.0/23			
10.0.2.0/24			
10.0.3.3 /25			
10.0.3.192/27			
10.0.3.244/30			

Table 1

CLO #3: Analyze the architectural principles of computer networking and compare different approaches to organizing networks.

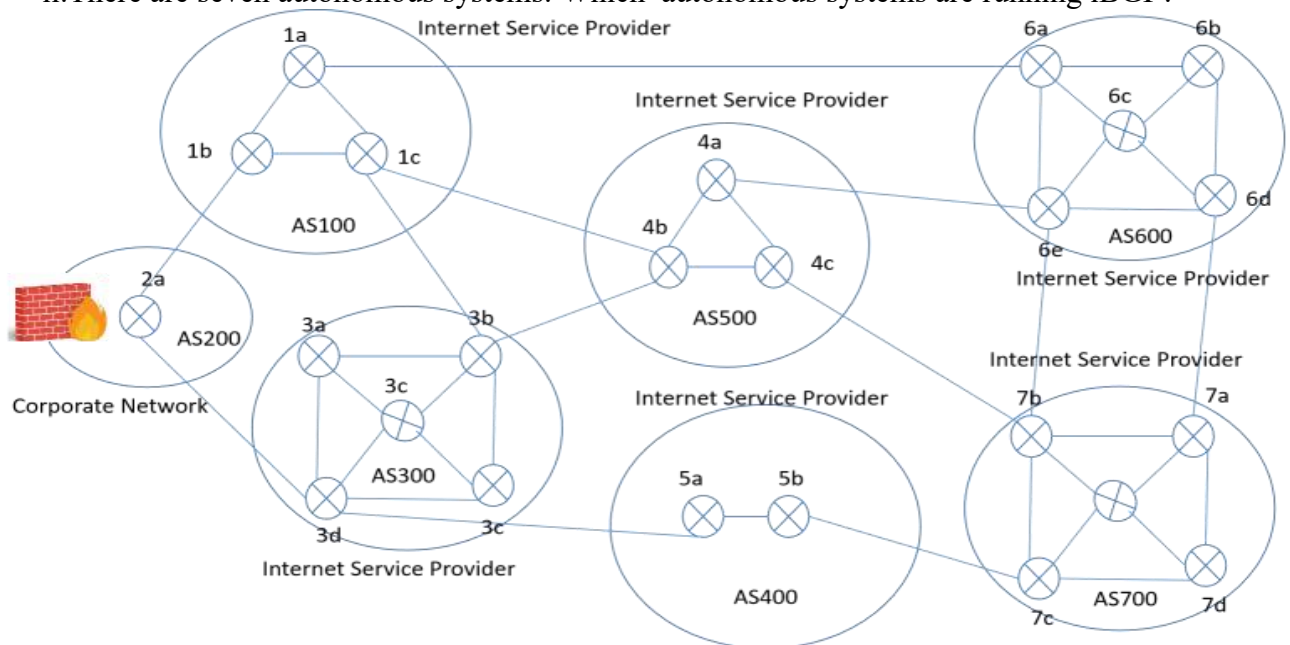
Q4. a) Using Dijkstra's algorithm, find the least cost path from source node **u** to all other destinations. Draw the resulting least-cost-path tree from **u**. **(10)**

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b) Consider the following exhibits running Border Gateway protocol. Answer the following (2.5+2.5):

- i. Identify the routers, which are designated as eBGP routers?
- ii. There are seven autonomous systems. Which autonomous systems are running iBGP?



CLO #3: Analyze the architectural principles of computer networking and compare different approaches to organizing networks.

Q5.a) What are the services provided by link layer? Describe briefly any two services. (5)

b) Describe channel partitioning methods along with their pros and cons. (5)

c) Consider the Cyclic Redundancy Check (CRC) algorithm. Suppose that the 4-bit generator (G) is 1001, that the data payload (D) is 101110 and $r=3$. What is the value of R? (5)